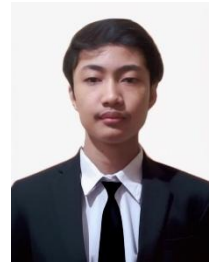


# Agil Herlambang Masadi

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Website Portfolio : <https://project-portfolio-agil-herlambang-m.vercel.app/>



## PROJECT AND WORK EXPERIENCE

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### Beca

Feb 2026 – Apr 2026

Role : Drainage Engineer - Freelance

Project : ELLE Resort and Beach Club Project

Work Scope :

- Conducted hydrological analysis to support the surface water drainage design for the ELLE Resort Project.
- Designed and modeled the stormwater drainage network, integrating flood mitigation structures such as retention ponds to manage the increased post-development runoff, strictly adhering to the Zero Delta Q principle.

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### PT. Jababeka

Feb 2026 – Apr 2026

Role : River Engineer - Freelance

Project : Design Review and Technical Recommendation for Binong and Sepatan River Diversion

Work Scope :

- Conducted hydrological analysis for the catchment areas encompassing the Cipagadungan, Cibinong, and Cisepatan river basins.
- Calibrated the flood discharge against the observed extreme flow data obtained from the Lemahabang Weir Gauging Station.
- Evaluated the performance of the existing retention pond in attenuating flood discharges from the Cibinong and Cisapetan rivers for 20, 25, and 50 year return periods.
- Performed low flow analysis to assess the dependable flow for the Cipegadungan, Binong, and Sepatan river basins.
- Executed hydraulic and sedimentation modeling to simulate various flood control scenarios, including flow diversions and retention ponds.
- Performed structural stability analysis for the retention ponds and channels under various water level scenarios (Low Water Level, Normal Water Level, and Flood Water Level), integrating seismic load variations for Operating Basis Earthquake (OBE) and Maximum Design Earthquake (MDE).

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### PT. Pasir Bara Prima

Dec 2025 – Mar 2026

Role : River Engineer - Freelance

Project : Study of the Kuap River Diversion

Work Scope :

- Delineating the Kuap Watershed (DAS).
- Conducting hydrological analysis for 5 rainfall stations and 1 water level station (Sei Kapuas Pujon).
- Performing hydraulic modeling of existing and diversion channel scenarios for the Kuap River diversion.
- Conducting sedimentation modeling for existing and diversion scenarios of the Kuap River to determine potential scour.
- Analyzed stabilization measures including stone masonry revetments at river bends, benched excavation slopes in highland areas, and clay embankments in lowland sections. Evaluated Factor of Safety under High Water Level (HWL), Low Water Level (LWL), and Rapid Drawdown conditions. Performed pseudo-static analysis for earthquake loads, considering both Operating Basis Earthquake (OBE) and Maximum Design Earthquake (MDE) scenarios for all conditions.

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## Self Employed

Nov 2025 – Mar 2026

Role : Water Resources Engineer - Freelance

Project : Planning of Embankment and River Normalization for Bojen River Flood Control in Bojen Village, Pandeglang Regency, Banten Province

Work Scope :

- Conducting hydrological analysis of the Bojen River watershed by calibrating ground rainfall stations with GPM satellite precipitation data. Conducting hydrological analysis for 5 rainfall stations and 1 water level station (Sei Kapuas Pujon).
- Performing hydraulic modeling of the Bojen River flood inundation triggered by a 25-year return period rainfall discharge.
- Designing the embankment and river normalization for the Bojen River to mitigate flood inundation, utilizing hydraulic modeling to optimize the design using HEC-RAS.
- Conducting stability analysis on the proposed embankment and river normalization designs using Geostudio.

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## Center for Infrastructure and Built Environment ITB, Bandung

Oct 2025 – Dec 2025

Role : Salt Irrigation and Coastal Engineer - Contract

Project : Integrated design and build construction work for development National Salt Industry Center Area (K-SIGN) in Rote Ndao Regency.

Work Scope :

- Designed multiple alternative seawater distribution schemes to establish a highly efficient supply system for each salt pond.
- Developed a comprehensive hydraulic model in HEC-RAS to evaluate and optimize the entire water system including inflow, spillway, and distribution schemes for a 1000.63 Ha salt production area.
- Executed hydrodynamic modeling of tides and significant wave heights for 50 and 100 year return periods to support the design of the seawater intake pipeline and jetty infrastructure.

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## Synergy Engineering, Jakarta

Aug 2025 – Oct 2025

Role : Drainage Engineer - Freelance

Project : Hydrology and Hydraulic Analysis for Kaliberau Dalam Field Development.

Work Scope :

- Conducted study the project area's of rainfall patterns and hydrological conditions to understand water-related risks and establish safe design criteria for all drainage infrastructure.
- Conducted design the drainage network for all critical areas, including the Central Processing Facility (CPF) and access roads, to ensure rainwater is effectively and safely managed.
- Conducted design the culverts under the main access roads, ensuring water can pass through safely without damaging the road structure.
- Conducted design energy dissipation structure at the downstream end of the culvert to reduce flow velocity prior to pond entry, thereby minimizing turbulence and preventing structural damage to the pond caused by scouring.
- Conducted design settlement pond to reduce the post development discharge to ensure the Zero Delta Q principle is satisfied.

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**DSDA (Dinas Sumber Daya Air) DKI Jakarta Province, Jakarta**

Aug 2025 – Nov 2025

Role : Assistant Hydrology Engineer - Contract

Project : Study of Water Balance for the Raw Water Potential of the Pesanggrahan, Angke, and Cengkareng Drain River Basins to Support the Revision of RISPAM and JAKSTRADA for SPAM Implementation in DKI Jakarta Province.

Work Scope :

- Conducted water quality data collection in Pesanggrahan, Angke, and Cengkareng Drain Watersheds (DAS).
- Collected secondary data in the form of topographic and land use spatial data.
- Performed data processing, hydrological and hydraulic analysis for the Pesanggrahan, Angke, and Cengkareng Drain Watersheds. This includes creating a rainfall runoff hydrological model, determining the duration curve to assess potential raw water sources for drinking water, and defining maintenance and environmental flows for the watersheds.
- Conducted water quality analysis for raw water supply needs in the Pesanggrahan, Angke, and Cengkareng Drain Watersheds based on the results of water quality testing.
- Provided engineering recommendations for increasing the dependable flow.

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**Bandung Institute of Technology, Bandung**

Aug 2025 – Dec 2025

*Research Assistant*

- Analyzed and mapped the historical changes of mangrove conditions along the DKI Jakarta coastline.
- Visualized historical erosion and sedimentation patterns in the coastal area of DKI Jakarta Province through spatial mapping.
- Identified, cataloged, and mapped the inventory of historical coastal protection structures to assess coastal defense evolution.
- Providing recommendations for critical areas suitable for mangrove planting to prevent erosion in the coastal area of DKI Jakarta Province.

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**PT. Atria Swacipta Rekayasa, Bandung**

Des 2024 – Feb 2025

Role : Drainage Engineer - Freelance

Project : Review Design and Recommendation Modification of Existing and New Surface Storm Water

## Drainage at Pangkalan Kerinci Mill Site, Town Site I & II

### Work Scope :

Conducted a comprehensive review of the existing drainage design for PT. Riau Andalan Pulp and Paper (RAPP) at Pangkalan Kerinci sites, providing recommendations for necessary actions such as sediment dredging and channel enlargement to enhance overall capacity. This process involved performing a detailed hydraulic analysis using PCSWMM software, which modeled 5, 10, and 25 year return period rainfall data to verify that the proposed improvements would ensure adequate capacity and prevent potential flooding.

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**SIBADES (Sipil Bangun Desa) HMS ITB, Bandung**

Aug 2024 – Dec 2024

Role : Hydrology and Hydraulic Engineer - Contract

Project : Evaluation of Scouring Bridge Construction in Bukanagara Village, Subang Regency.

### Work Scope :

- Conducted a hydrological analysis to calculate the 25 year flood discharge (Q25) of the Cikaruncang River for a proposed bridge project in Bukanagara Village, Subang Regency.
- Performed hydraulic modeling using HEC-RAS to determine the flood water levels and inundation area of the Cikaruncang River for the 25 year return period event.
- Determined the optimal bridge span and deck elevation to prevent massive local scour and ensure adequate freeboard above the 25 year flood (Q25) level. This analysis was performed by utilizing the bridge modeling and local scouring features within HEC-RAS. Validating the results from HEC-RAS by comparing them against manual scour depth calculation.
- Developed alternative designs for scour protection measures at bridge piers and abutments.

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**Ministry of Public Works and Public Housing, Trenggalek**

Jun 2024 – Jul 2024

Role : Internship DAM Engineer

Project : Bagong Dam Construction Project Package 1 in Trenggalek Regency (Multi-Year Contract)

### Work Scope :

- Delineated the Bagong Dam Watershed and collected secondary data, including rainfall, soil type, and land cover, to support the hydrological analysis.
- Conducted a hydrological analysis to determine the flood discharge and dependable flow generated from the Bagong Dam Watershed.
- Calculated the maximum release discharge from the Bagong Dam to establish an annual reservoir operation pattern, ensuring the water level returns to its normal elevation and does not fall below the intake level.
- Conducted a stability analysis of the Bagong Dam for various failure scenarios in accordance with SNI 8460-2017, utilizing Geostudio software.
- Analyzed and executed the filling and compaction processes for the Bagong main dam embankment.
- Analyzed and executed the grouting process for the Bagong main dam.

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**Bandung Institute of Technology, Bandung**

*Assistant Lecturer*

- Assistant Lecturer of River, Swamp, and Lake Ecosystems
- Assistant Lecturer of Coastal Engineering
- Assistant Lecturer of Technical Drawing
- Assistant Lecturer of Soil Mechanics

- Assistant Lecturer of Introduction to Engineering and Design

## EDUCATION

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**Bandung Institute of Technology**, Bandung Aug 2021 – Aug 2025

*Bachelor of Engineering, Water Resources Engineering and Management*

- Final Project : Modeling and Mitigation of Earthquake-Induced Tsunami in the Coastal Area of Gianyar Sub-district, Bali Province.
- GPA 3.59 of 4.00
- Honors: Cumlaude

**Bandung Institute of Technology**, Bandung Aug 2024 – Aug 2026

*Master of Engineering, Water Resources Management (Fast Track Program)*

- Thesis Project : Assesment of the Impact Tsunami Induced Sediment Transport on the Coastal Area and Tsunami Embankment in Gianyar Sub-District, Bali Province.
- GPA 3.90 of 4.00

**Member of KelasFullstack by Codepolitan** Aug 2022 – Aug 2024

## ACHIEVEMENTS

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**1st Place Winner of CINNERTION (Civil Innovation Paper Competition) College 13th Edition at ITS, Surabaya**

**Finalist AGRIFASCO (Agricultural Farming System Competition) 9th Edition at ITB, Bandung**

**Finalist LOGRAK (Lomba Gagasan dan Rancangan Kreatif) 2024 at POLINEMA, Malang**

**TOP 15 Participant KBAI (Kompetisi Bangunan Air Indonesia) 2025 at UB, Malang**

**2nd Best KKN Group, Theme of Energy**

**Ganesha Talent Assistantship Scholarship (GTA-100) Awardee at ITB, Bandung**

**Ganesha Karya Awardee at ITB, Bandung.**

## LEADERSHIP ACTIVITIES

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**HIMASDA (Himpunan Mahasiswa Sumber Daya Air) ITB, Bandung** June 2024 – Des 2024

*Head of Fundraising*

- Led fundraising initiatives to support the impactful programs of HIMASDA ITB through partnership programs and e-commerce sales.
- Developed and managed an e-commerce platform for HIMASDA ITB online sales activities.

**KKPA (Kuliah Kerja Penyelamatan Air) 2023, Sumedang** Nov 2023 – Dec 2023

*Head of Pipanization*

- Conducting a survey related to the construction program of a small dam, and piped water supply in Dusun Cikaso, Desa Cimarga, Kecamatan Cicitu, Kabupaten Sumedang.
- Supervising and implementing the progress of the piped water program to address the clean water crisis issue in Dusun Cikaso

**KKN (Kuliah Kerja Nyata) ITB 2023, Sumedang** May 2023 – Aug 2023

*Head of Program*

- Led the strategic planning for a 12 member energy group's Community Service Program (KKN), which focused on providing public street lighting in Dusun Cikaso, Desa Cimarga, Kecamatan Cicitu, Kabupaten Sumedang.
- Headed the implementation process for the installation of public streetlights and delineators, successfully

installing a total of 11 street light units and 30 delineators.

## CERTIFICATIONS

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Analisis Hidrologi Software Hec HMS</b> , Kursus Sipil                                                                                                                 | Mar 2021            |
| <b>Aplikasi Geostudio Geotechnical Engineering - Analisis Stabilitas Bendungan</b> , Kursus Sipil                                                                         | Feb 2022            |
| <b>Extreme Wave Modelling Workshop (Delft-3D Wave) (Study Case : Breakwater)</b> , Kursus Sipil                                                                           | Jun 2024            |
| <b>Intensive Course: Pembuatan Model Alternatif Solusi Pengendalian Banjir, Time construction, serta Perhitungan Roughness Cost dengan SWMM</b> , Kursus Sipil            | Jun 2024            |
| <b>Kajian Penanganan Longsoran dan Rembesan di Hilir Bendungan dengan Metode Analisa Balik menggunakan Bantuan Software Geostudio Sesuai SNI 8460-2017</b> , Kursus Sipil | Jul 2023            |
| <b>Pelatihan Geoteknik Eksklusif (Beginner Level) dengan Bantuan Software Plaxis 2D V20 (8 Study Case Complete Package)</b> , Kursus Sipil                                | Oct 2024 – Nov 2024 |
| <b>Perancangan Struktur Bangunan Tahan Gempa "Irregular Building" &amp; pengecekan Ketidakberaturan dengan software ETABS berdasarkan SNI 1727-2020</b> , Kursus Sipil    | Sep 2024            |
| <b>Python for Data Professional Beginner - Part 1</b> , DQLab                                                                                                             | May 2023            |
| <b>Python for Data Professional Beginner - Part 2</b> , DQLab                                                                                                             | May 2023            |
| <b>Python for Data Professional Beginner - Part 3</b> , DQLab                                                                                                             | May 2023            |
| <b>A Practical Introduction to 2D River Modelling in HEC-RAS</b> , Udemy                                                                                                  | Jul 2025            |

## LANGUAGES AND SKILLS

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- Bahasa Indonesia (Native), English (Intermediate).
- MS Word, MS Excel
- Python, HTML, CSS, JS, CSS Framework Bootstrap.
- HEC-RAS, HEC-HMS, Delft3D, Geostudio, Plaxis, QGIS, ArcGIS Pro, Global Mapper, EPA SWMM, AutoCAD, EPANET, WEAP (Water Evaluation and Adaptation Planning), ETABS, PCSWMM, MODFLOW6, iRIC.